Emma Steyn

Bachelor of Architectural Studies

Portfolio of Selected works: 2024-2026

About me

With a strong foundation in design, research and problem-solving, I bring energy, optimism and curiosity into every project. I am eager to apply my developing skills, to learn from professionals and contribute meaningfully to your practice during an internship.

Education
School of Explorative Architecture
Bachelor of Architectural Studies
2024-2026
Average of High Pass (>70%) across all years

Bachelor of Science in Human Life Sciences
Stellenbosch University
2021-2023
Dean's Merit List Student

Experience
SAOTA
Student Intern
June - July 2026

Hours Clear Architects
Student Intern
September - November 2025

Language
English
Afrikaans

Contact
0768556481
emmasteyn4@icloud.com

Skills
Revit
Lumion
Rhino
Twinmotion
Procreate
AutoCAD
Affinity (Similar to Adobe suite)
3D printing
Laser Cutting

Qualities
Attention to detail
Clear communicator, with strong teamwork and collaboration skills
Reliable, organised and able to perform well under pressure
Positive and proactive mindset, eager to learn.

Achievements
New Architecture Now 2025 Finalist
Recognised for an innovative design proposal, selected among top entries, My work formed part of a curated exhibition at the AZA Conference.

Contents:

House Kamter
Red Box Theatre
Seriti
Light
Woodstock Furnace
Gathered Voices
Crèche
Op-o-site

House Kamter

emerging from the mountains

Project Type Residential

Location Geirangerfjord, Norway

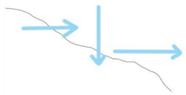
Year 2026

Software Revit
Lumion
Procreate
AI Rendering Software

About Conceived as an extension of the surrounding mountains. A solid concrete lower level anchors the home firmly to the terrain, while a lighter upper volume, clad in charred timber, rises above it. The contrast between the two materials creates a sense of permanence and lightness, allowing the building to appear as though it is emerging naturally from the landscape.

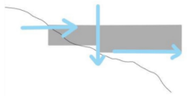
Existing Topography

The existing topography of the site establishes the primary design constraint and becomes the foundation from which the architectural form is developed.



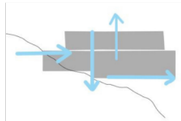
Shaped by the Fjord

The building footprint extends along the contours of the mountain, minimizing intervention while reinforcing a strong connection to the terrain.



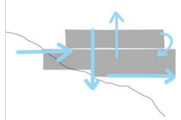
Embedding the Lower Level

The lower volume is carefully integrated into the slope, allowing the architecture to emerge naturally from the landscape rather than sit above it.



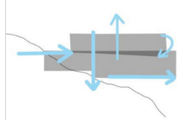
Elevating for Light and Views

The upper level is introduced to capture panoramic fjord views while increasing access to natural daylight and passive solar gain.



Architectural Language

The upper volume mirrors the footprint of the lower level, establishing a cohesive architectural language.

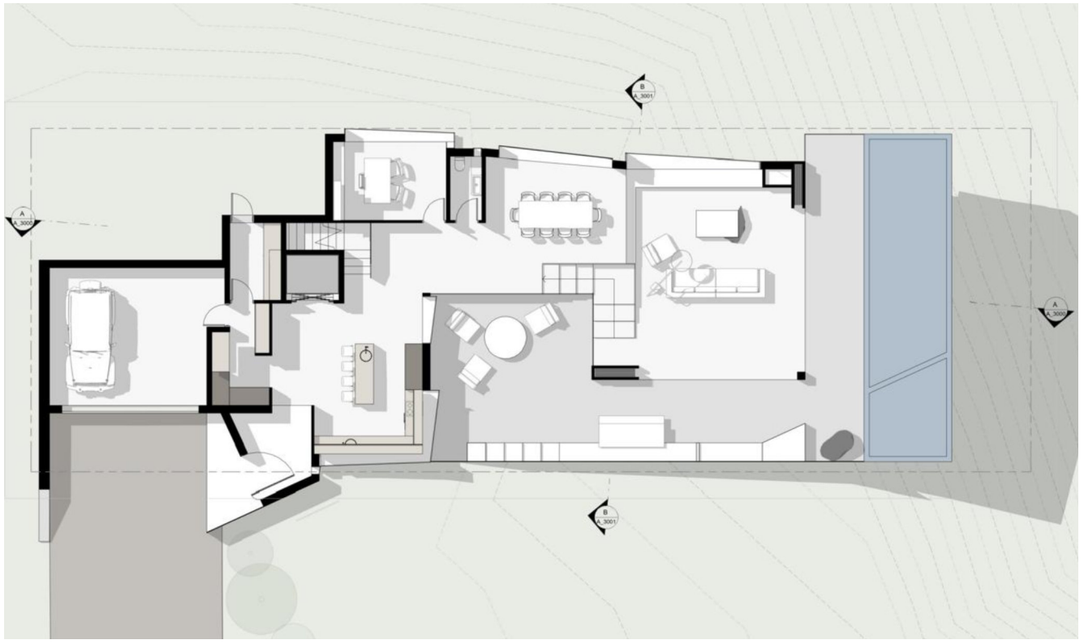


Orienting toward the Fjord

The upper floor is rotated toward the fjord, creating a more dynamic relationship with the surrounding environment.

Material Palette

Charred Timber
Concrete
Blackened Steel



Ground Floor Plan

First Floor Plan

Main Bedroom

Upstairs Hallway

Kitchen

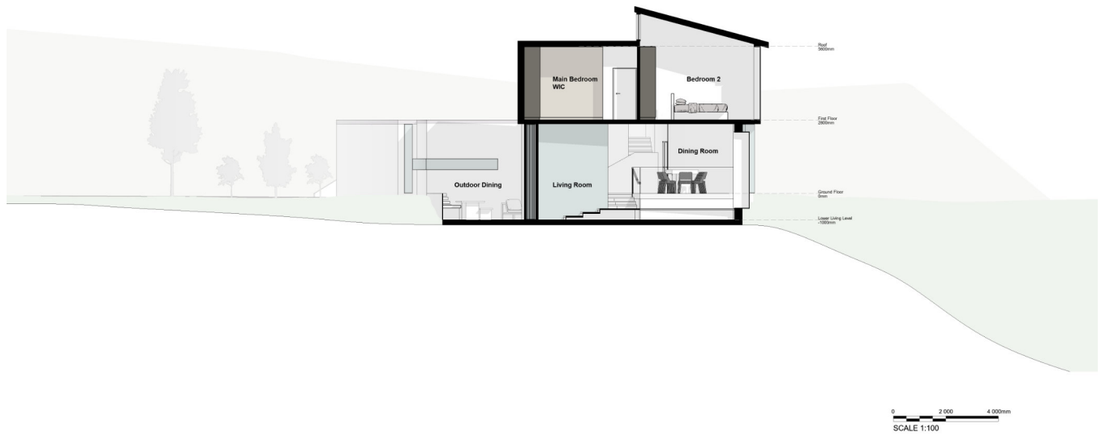
Pool + BBQ Area

Exterior

BBQ area

Section AA

0 2 000 4 000mm
SCALE 1:100



0 2 000 4 000mm
SCALE 1:100

Section BB

Red Box Theatre

city as a stage

Project Type Mixed-use civic building
Theatre, Gallery and Drama School

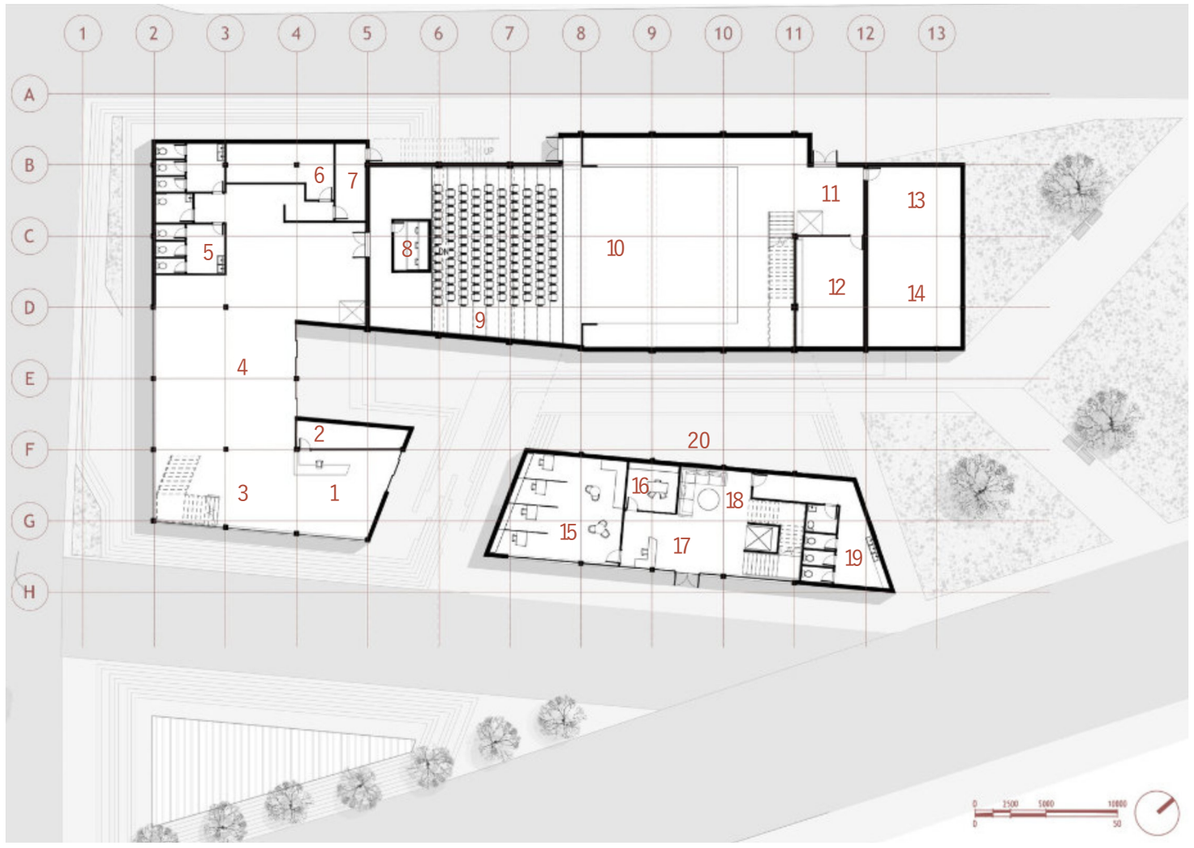
Location 10 Tuin Plein Street, Cape Town

Year 2026

Software Revit
Lumion
Procreate

About To the idea of city as a stage, where everyday life becomes the performance. My building consists of a reinforced concrete frame positioned to frame movement and views across the site. One volume engages with the public realm by exposing rehearsal spaces to the street, while the other offers a cafe oriented toward the landscape. Through the integrated ramps and inhabitable thresholds, the building dissolves the boundary between audience and performer, embedding theatre in everyday life.

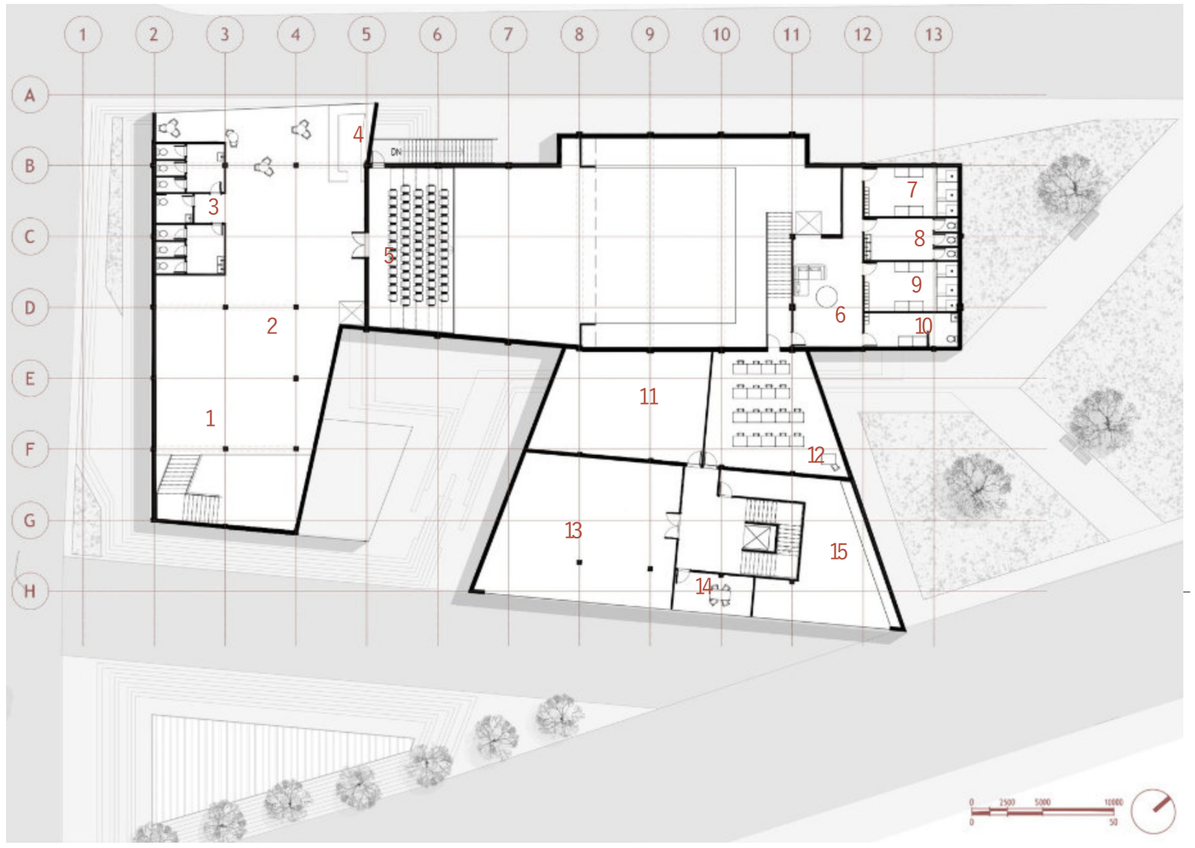
Ground Floor Plan



Stage BOH, Education Foyer + Gallery Foyer

1.Tickets/Info
2.Cloakroom
3.Foyer + Public Hall
4.Gallery
5.WC
6.Storage
7.Waste + Recycling
8.Control Booth
9.Auditorium
10.Stage
11.Scene Dock/Loading Prep + Workshop
12.Props/Set Storage
13.Plant Room
14.Elec/IT
15.Staff Offices
16.Head Office
17.Reception
18.Student Break Room
19.Education WC
20. Outdoor Amphitheatre

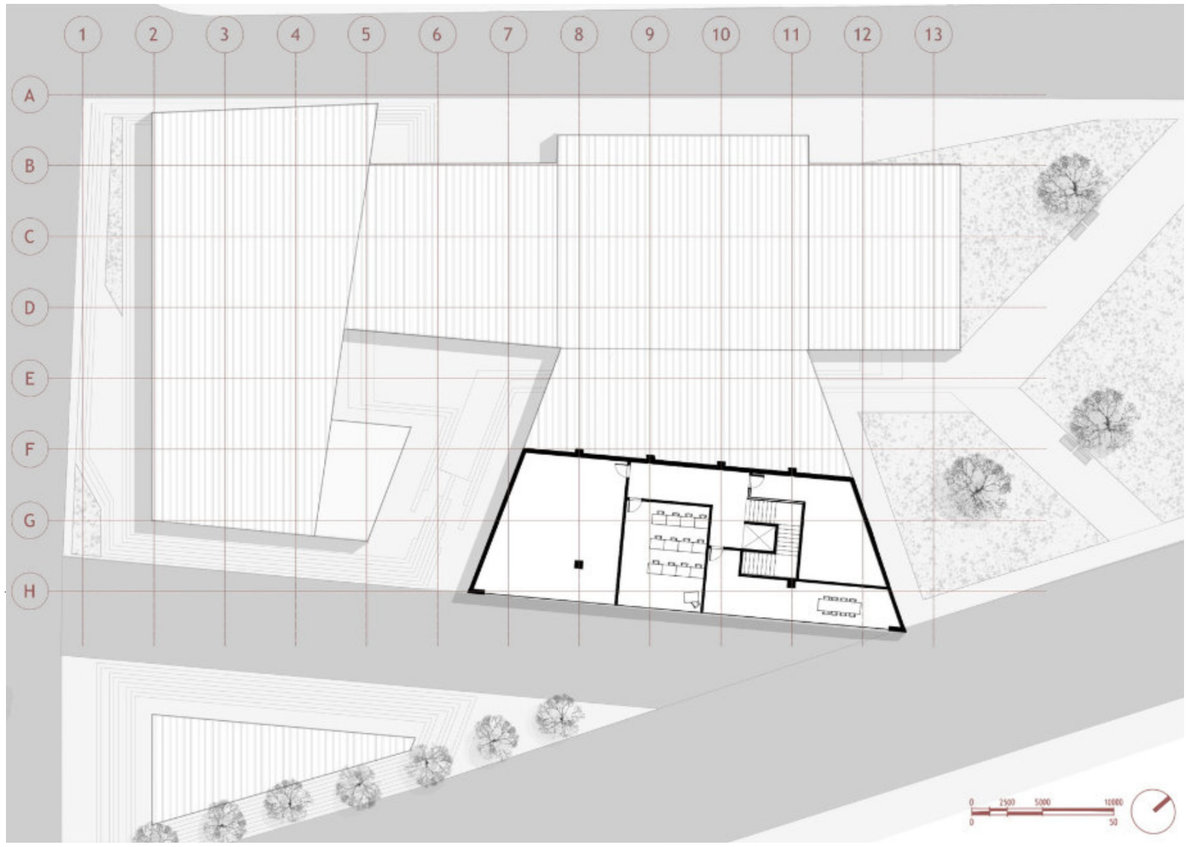
First Floor Plan



Stage BOH, Education Classrooms +Upper Gallery

1.Upper Foyer
2.Upper Gallery
3.WC
4.Cafe
5.Upper Auditorium/Gallery Seating
6.Green Room
7.Dressing Room 2
8.Showers
9.Dressing Room 1
10.Accessible WC + Dressing Room
11.Studio 1
12.Class 1+ Link from Education to Theatre
13.Rehearsal Studio/Movement Room
14.Meeting Room
15.Resource/Media Room

Second Floor Plan



Education Classrooms

1.Studio 2
2.Class 2
3.Storage
4.Seminar/Crit Room

Cafe

This space, with it's floor to ceiling, west-facing windows, frames beautiful views of lions head and the upper gardens/tamboerskloof area. This space operates as a cafe during the day and as an space where theatre visitors can have a quick drink during interval.

Theatre

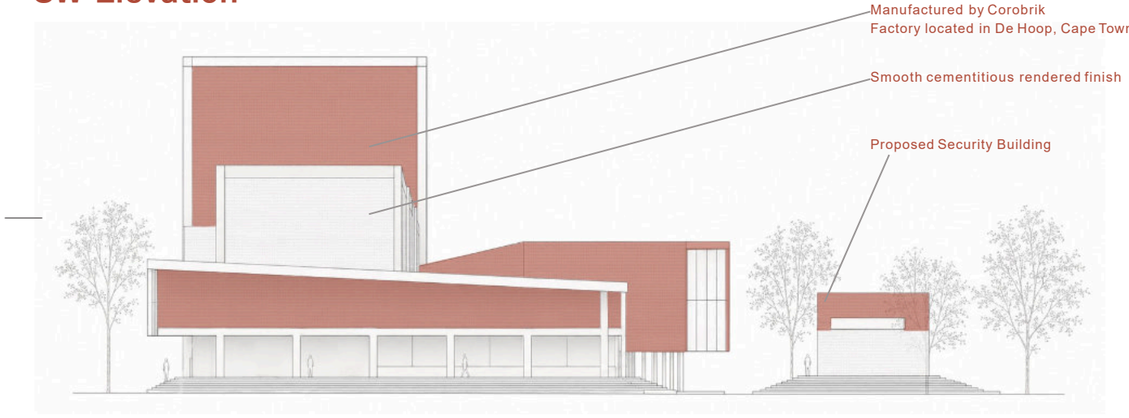
Featuring 200 seats, a 120sqm stage, wheelchair accessible seating, a control booth and controlled access to the lights/gantry.

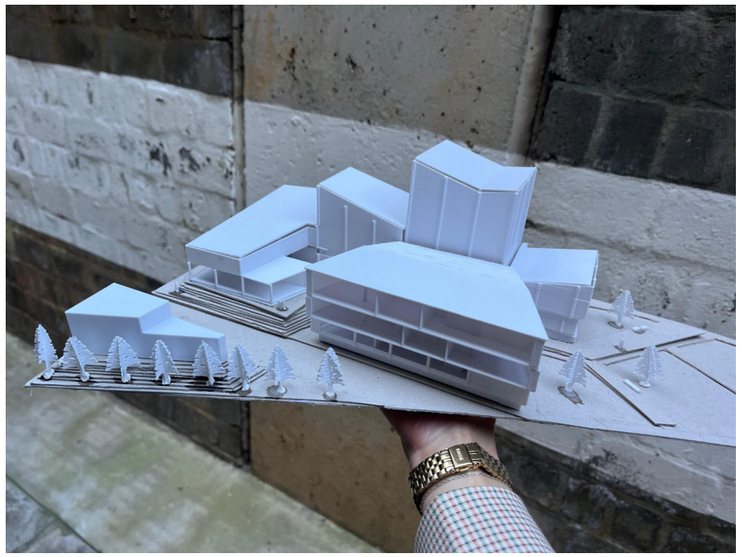
The perfect space for a range of events.

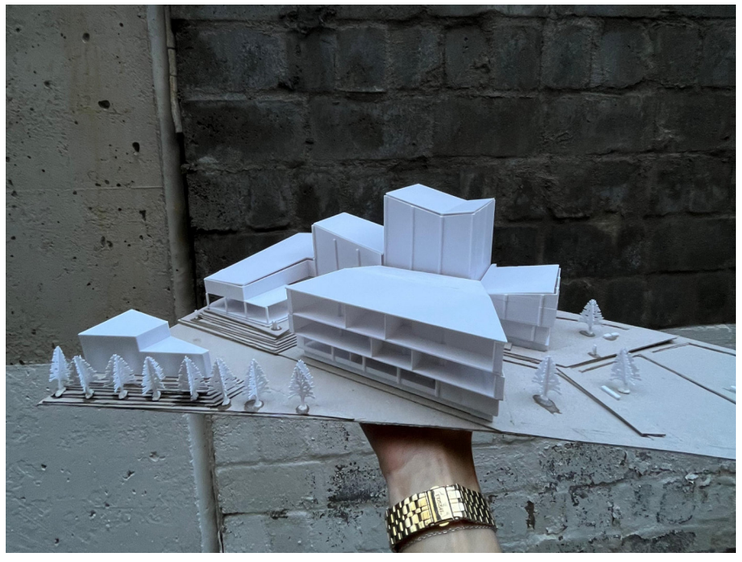
Foyer

This is the main point of arrival for gallery and theatre guests. Featuring a statement staircase, and a cloakroom to the right.

SW Elevation







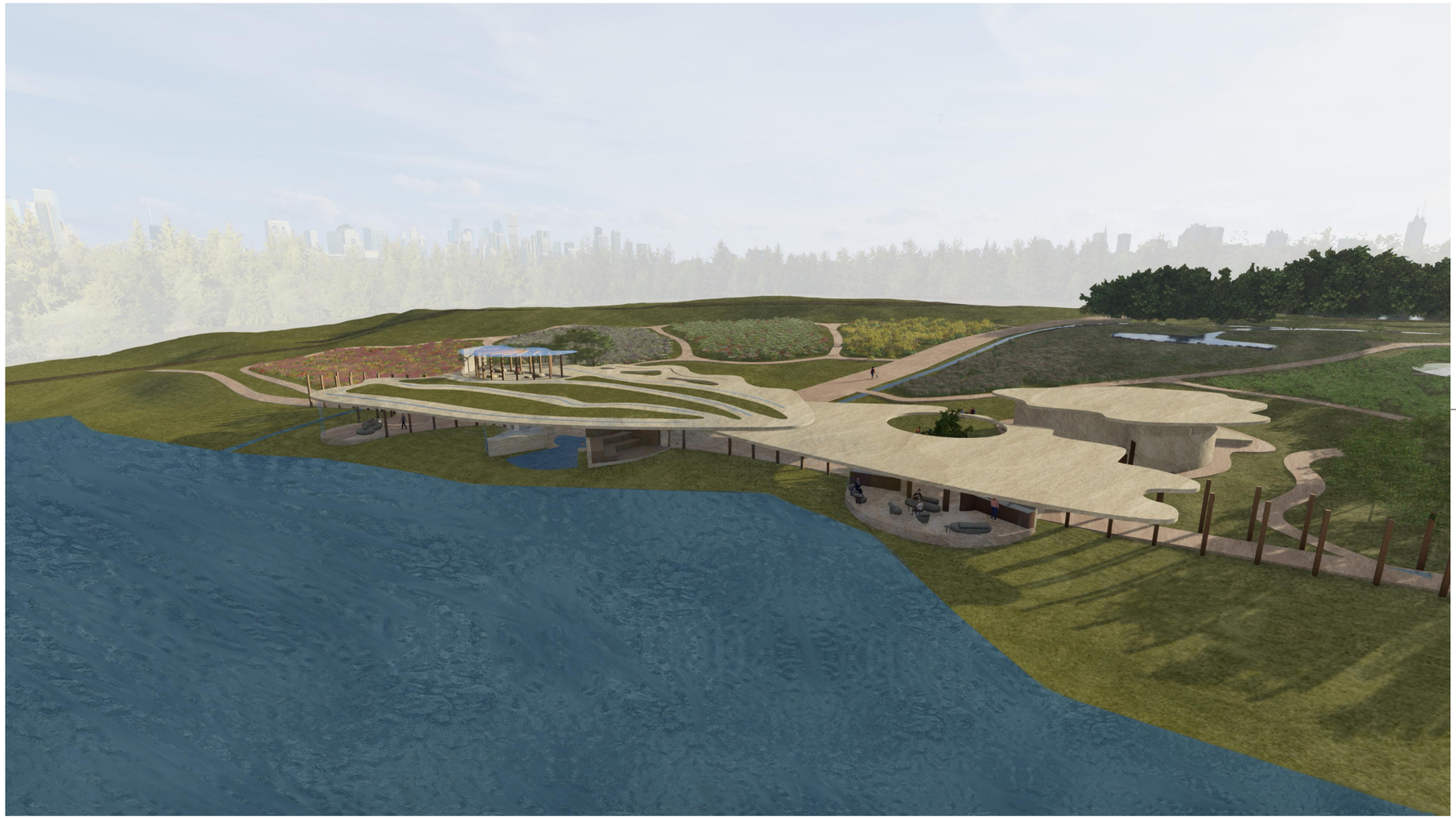
Physical Model 3D Printed, creating a minimal Revit model, exporting as an STL file and printed using the Bambu Lab A1 mini

Scale 1:200

View from Hope Street

Seriti

connection, renewal, belonging



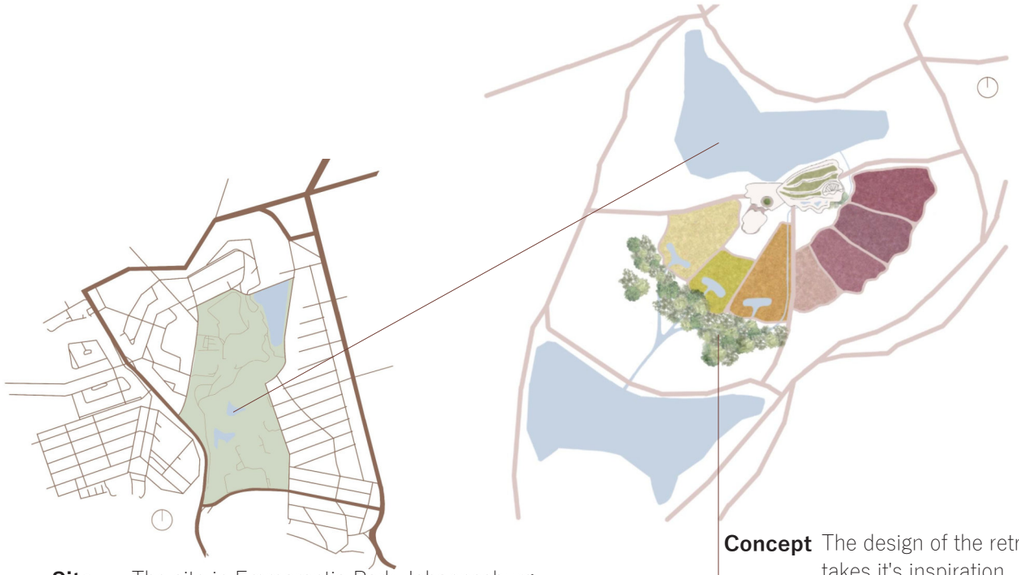
Project Type Retreat in a Public Park

Location Emmerentia Park, Johannesburg

Year 2025

Software Revit
Lumion
Procreate
Affinity

About The brief was to design a retreat centered around the four senses: sound, taste, touch and sight. Over four weeks, it developed into a multi-sensory building proposal that pushed me to work experimentally and expand my digital skills.



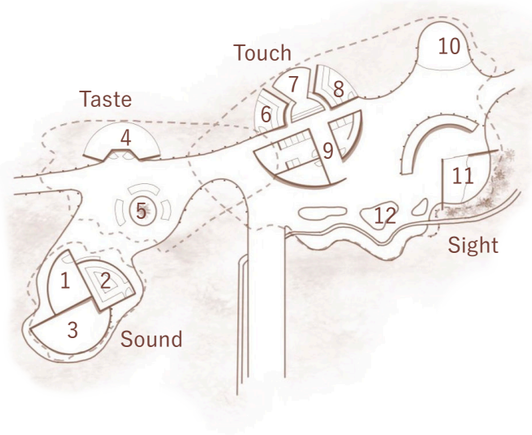
Site The site in Emmarentia Park, Johannesburg, between the two smaller dams, offers a rich landscape shaped by water and vegetation.

Concept The design of the retreat takes it's inspiration from the brown-veined butterfly, whose wings are reflected in the shape of the roof.

Each year this butterfly migrates from the Karoo region across southern Africa to Madagascar, often passing through the Gauteng area.

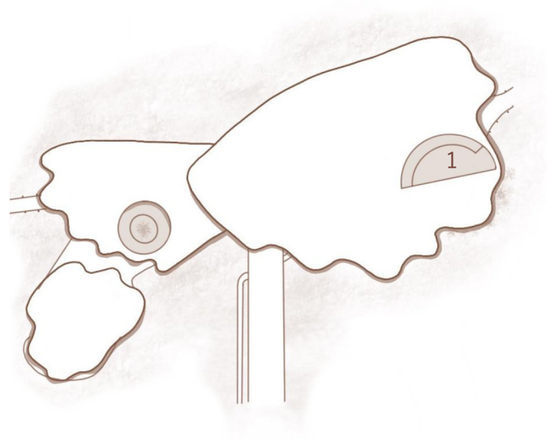
Intervention Surrounding the retreat, the Shepherd's Tree (Boscia albitrunca) will be planted. This tree is a host plant from the butterfly, providing shelter along it's journey.

By creating a landscape filled with these trees, the retreat not only attracts and supports these migrating butterflies for longer stops, but also enriches the local environment,bringing in more wildlife, strengthening biodiversity, and weaving the buildings into the ecosystem of Emmarentia Park.



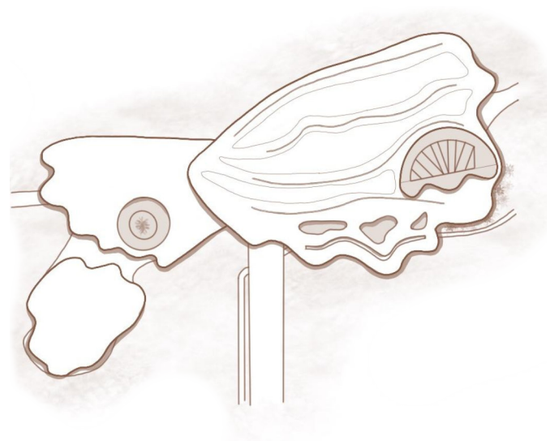
Ground Floor Plan

1. Echo Chamber
2. Rain Listening Area
3. Amphitheatre
4. Tea Tasting Room
5. Picnic Garden
6. Sauna
7. Plunge Pool
8. Steam Room
9 WC
10. Entrance
11. Bird Watching Room
12. Reflection Garden



First Floor Plan

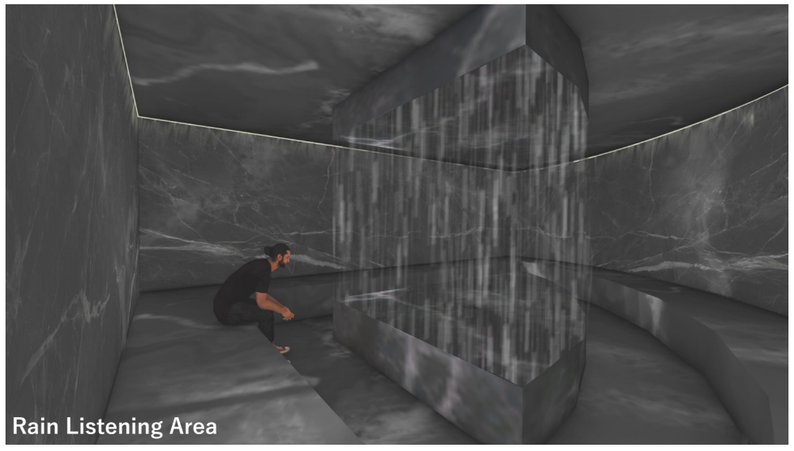
1. Stargazing Court



Roof Plan

Wildflower Garden

Picnic Garden



Rain Listening Area

Tea Tasting Room

Light as Design Informant

short course exploring abstract ideas as design informants

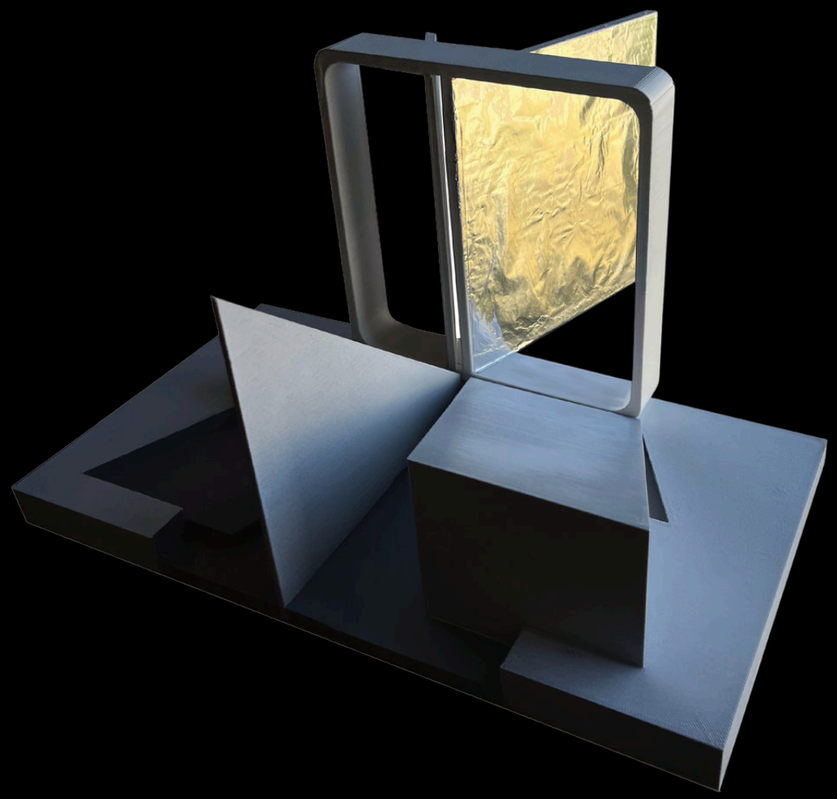
Elective How can an Abstract Architectural Value act as a Design Informant

Topic I chose to research how light can act as a design informant.

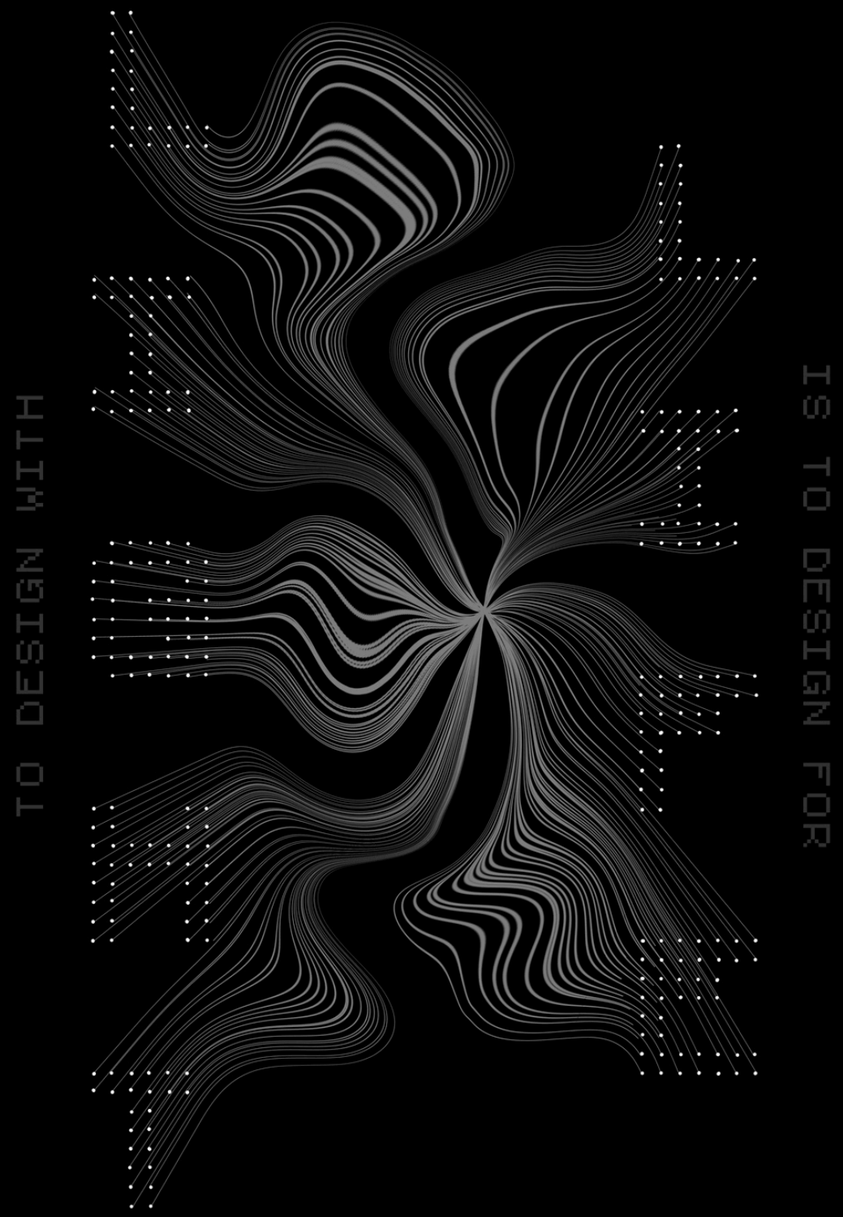
Year 2026

Software Procreate
Affinity

About Although this project began as a highly abstract exploration, the final hand-in required us to translate our ideas into an architectural application. In response, I developed a series of light modules designed to harvest and redirect natural light into a building at different times of the day. By controlling the amount of direct sunlight and reflected light entering a space, these modules create changing light conditions that respond to the body's natural circadian rhythm, supporting human health and well-being.



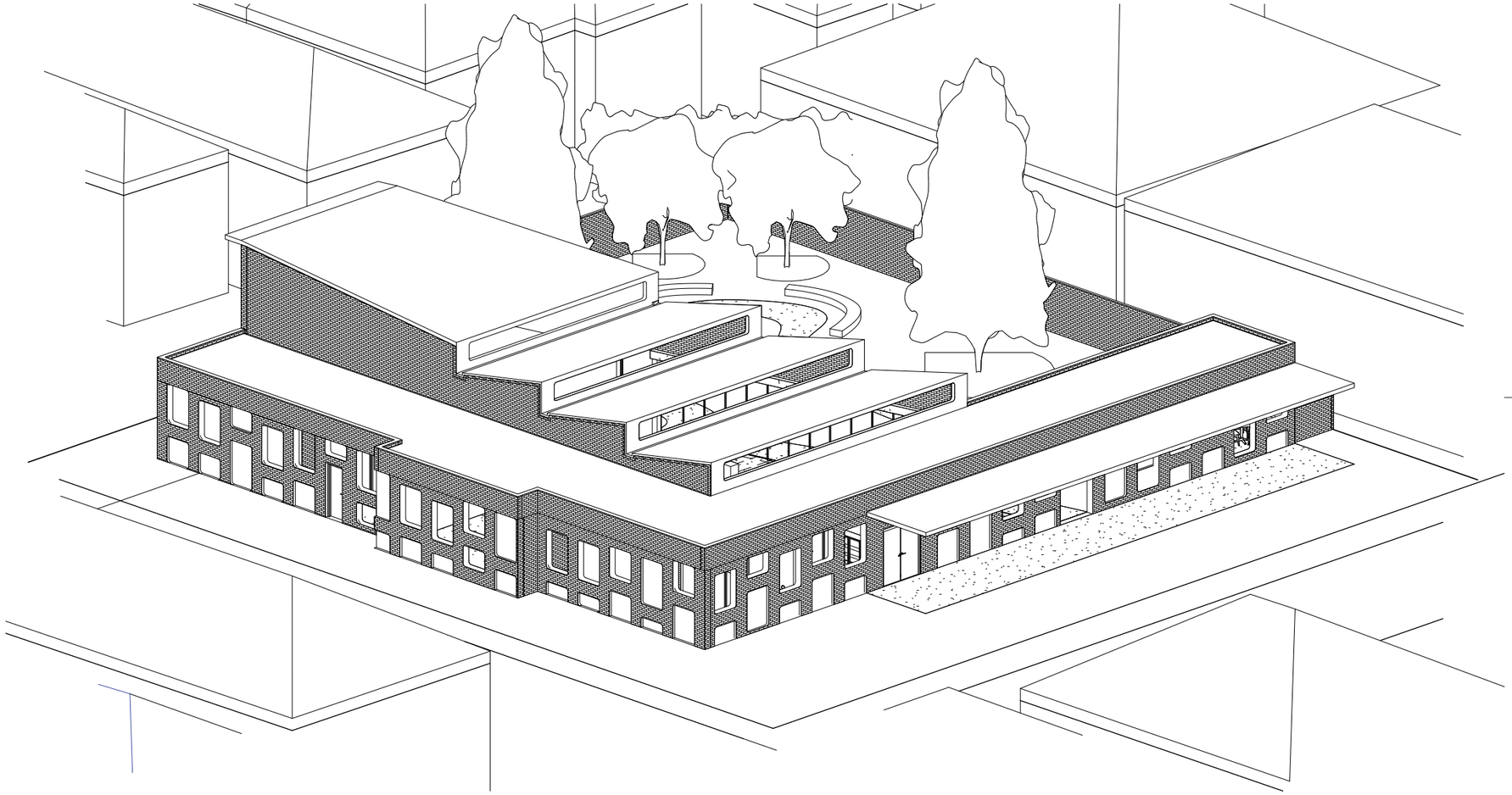
Conceptual Model Exploring the shadows and light manipulated by the reflective light module



Poster Design to design with light, is to design for life

The Woodstock Furnace

architecture to preserve history



Project Type Glass Blowing Workshop, Gallery and Coffee Shop

Location Corner of Station Rd and Arnold St, Observatory

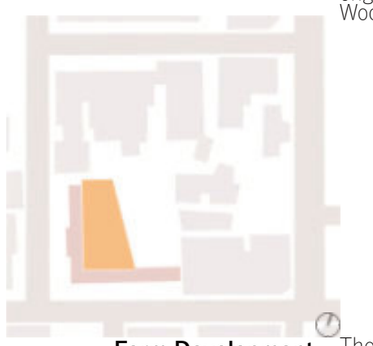
Year 2025

Software Revit
AutoCAD
Procreate

About This project is inspired by South Africa's first glass factory, which operated in Woodstock from 1878 to 1882. Through a memorial garden, historic mosaic patterns, and a form derived from the original site footprint, the project uses architecture to preserve history.

Sun Analysis

Wind Analysis

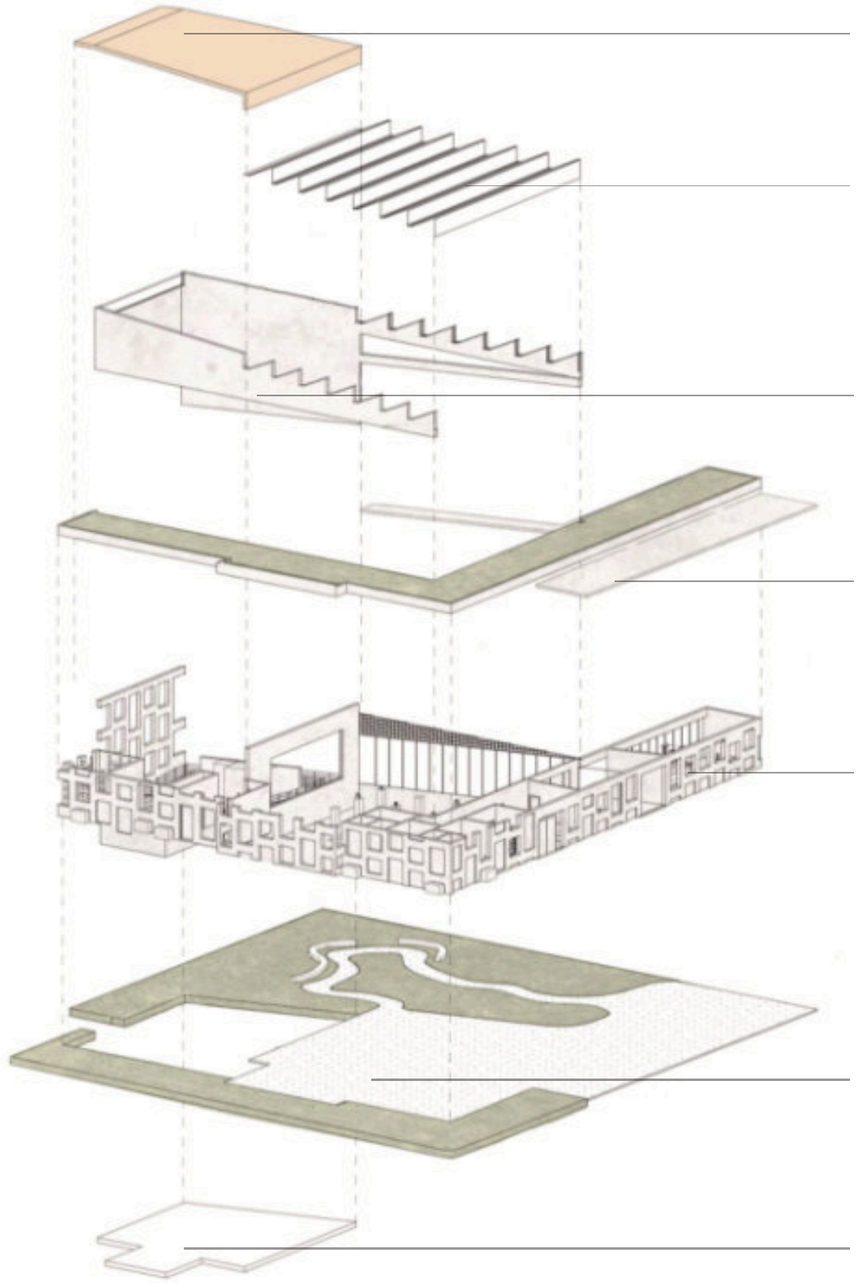


Form Development The form of the primary spaces, the gallery and the workshop, originate from the footprint of the Woodstock Glass Factory site.

Form Development The auxiliary spaces wrap around the primary volume in an L-shaped form, providing the necessary service and support areas.

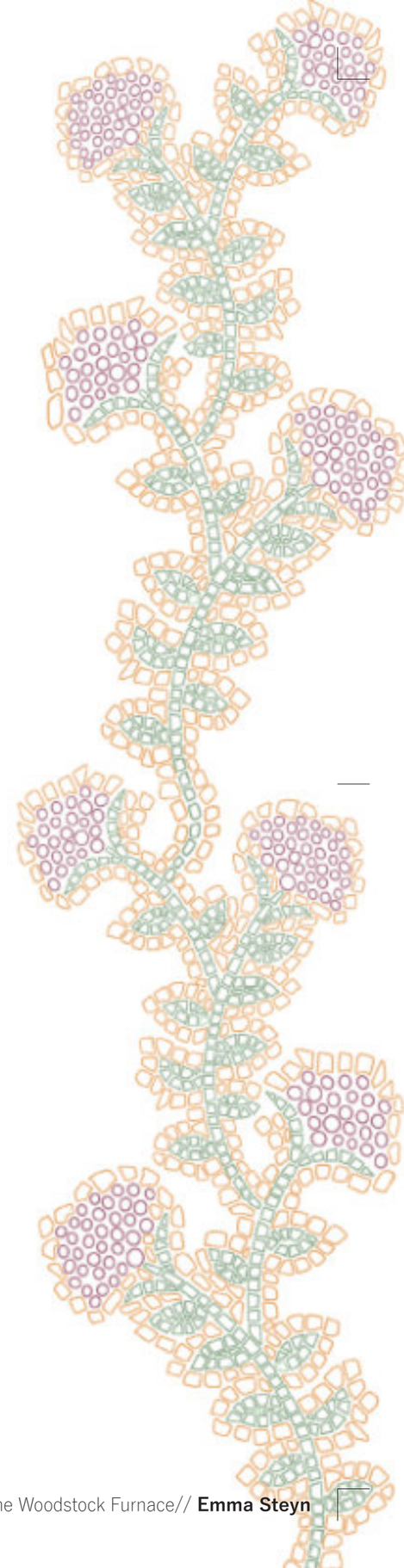
1. Delivery Entrance
2. Storage
3. Sunken Workshop
4. Staff Bathroom
5. Staff Entrance
6. Staff Kitchen
7. Gallery
8. Women's Bathroom
9. Wheelchair Accessible Bathroom
10. Men's Bathroom
11. Office
12. Entrance
13. Meeting Room
14. Key-holder entrance to park
15. Cafe
16. Memory Garden + Seating

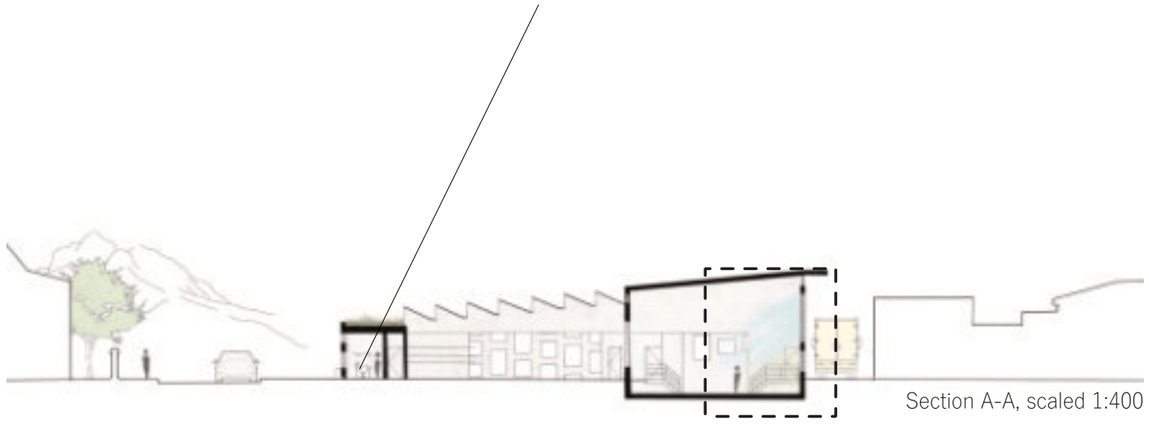
1:200



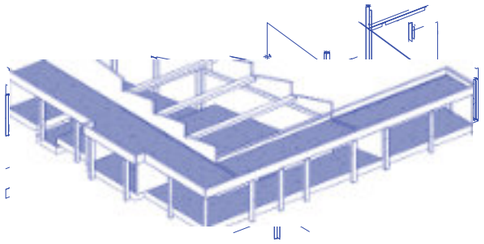
Memory

The mosaic detail on the benches in the garden is adapted from a historic pattern originally found on Woodstock Glass Factory products.
The tiles will be hand-laid by a local craftsman.



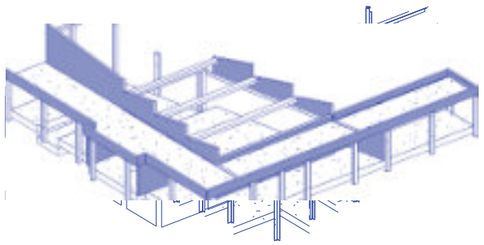


Section A-A, scaled 1:400



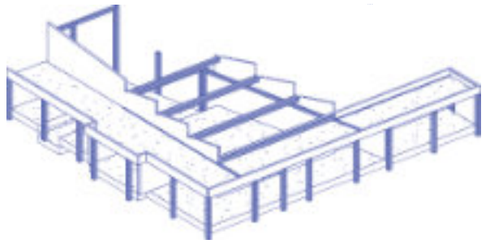
Floors

+

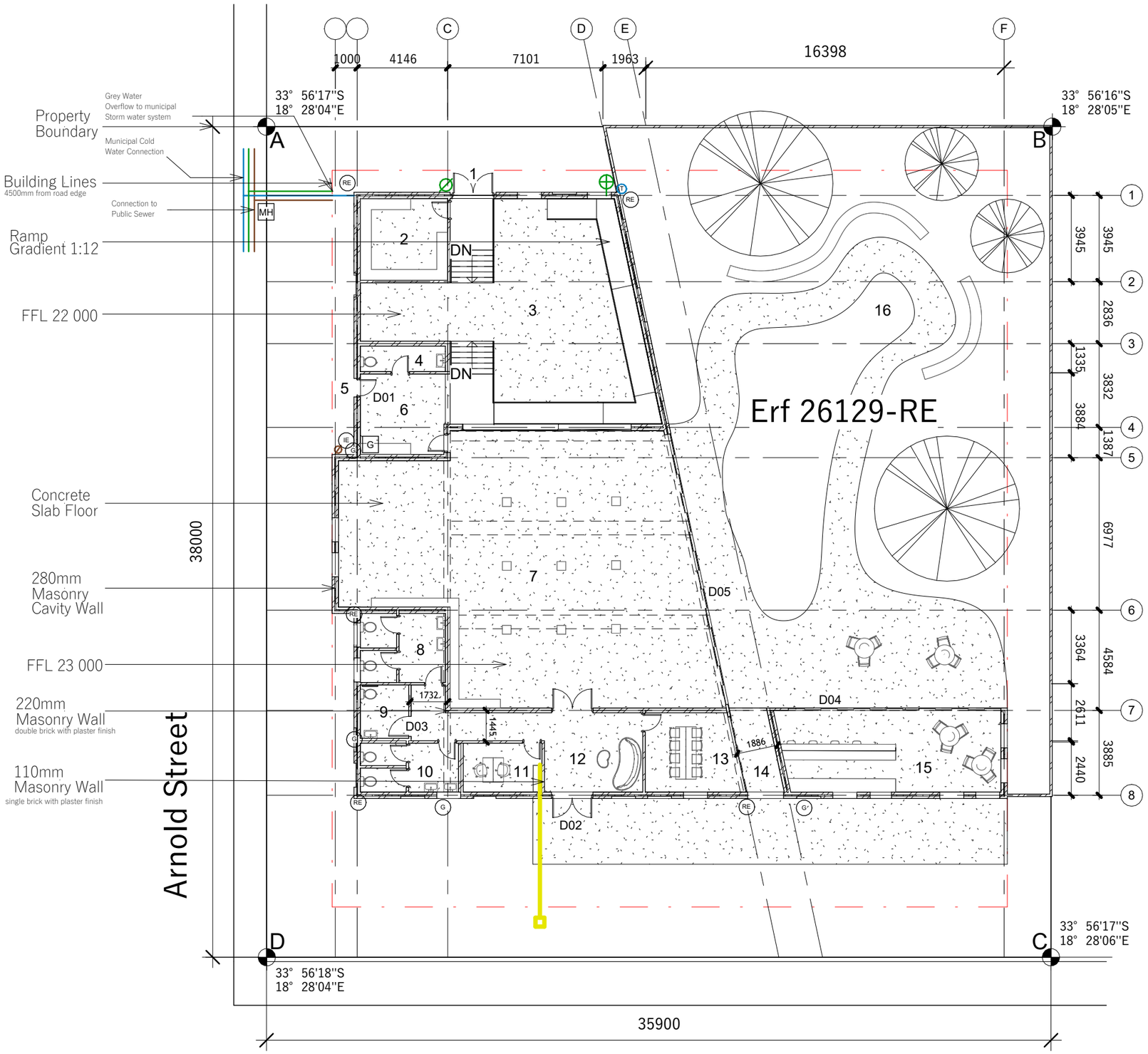


Walls

+

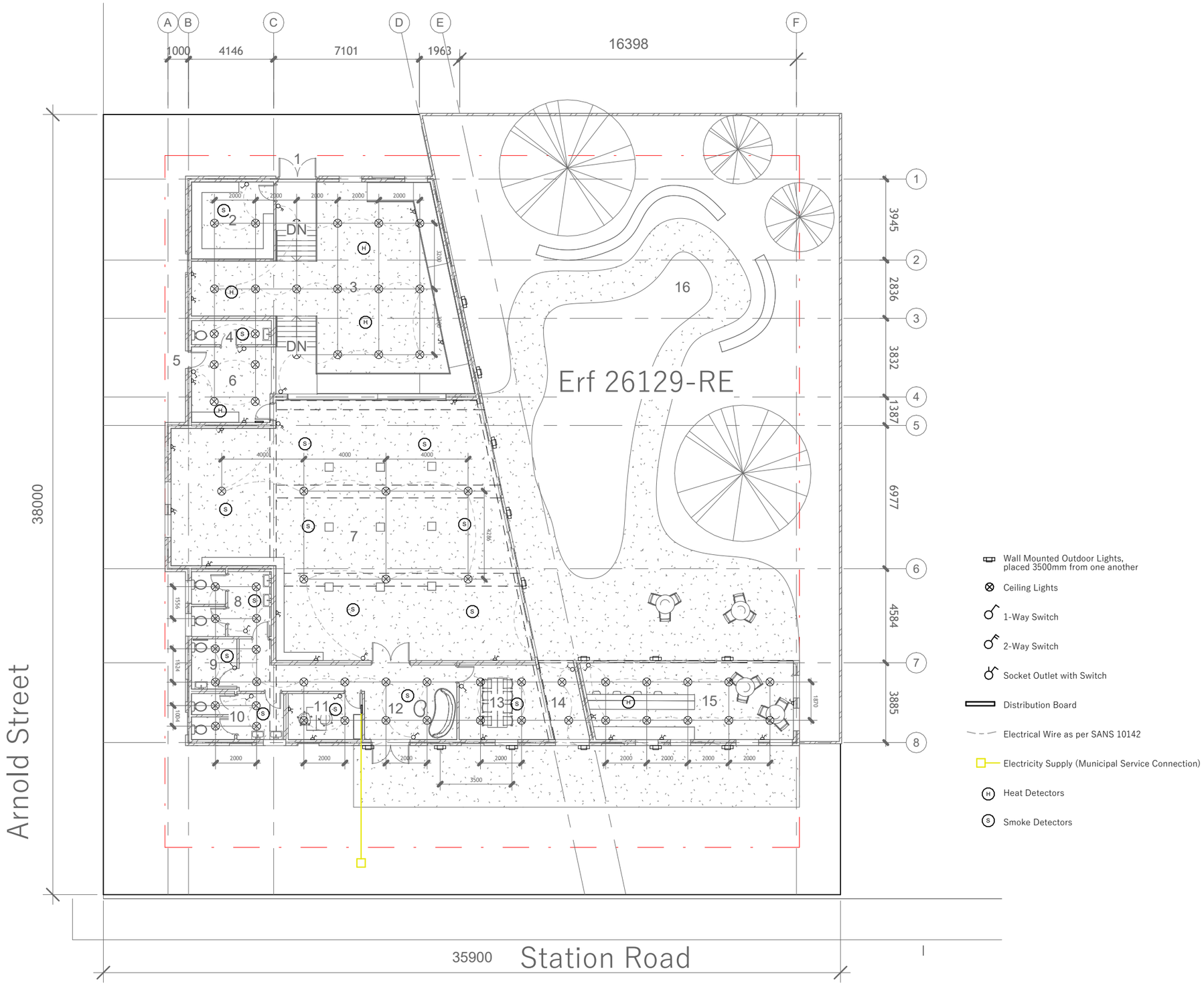


Beams & Columns



1 Delivery Entrance
2 Storage - 15sqm
3 Workshop - 115sqm
4 Staff Bathroom - 5sqm
5 Staff Entrance
6 Staff Kitchenette
7 Gallery Space - 180sqm
8 Women's Bathroom - 12 sqm
9 WC Bathroom - 6 sqm
10 Men's Bathroom - 10 sqm
11 Office - 8sqm
12 Entrance - 26sqm
13 Meeting Space - 15sqm
14 Garden Entrance
15 Cafe - 38 sqm
16 Memorial Garden

- Gutter Down Pipes 100mm
- 5000L Grey Water Tank
  Filters Water used for Garden Irrigation
  Overflow connected to municipal sewer
- Rodding Eye
- Inspection Eye
- Man Hole
- Gullies
- Electricity Supply
  (Municipal Service Connection)
- Tap
- 200L Geyser



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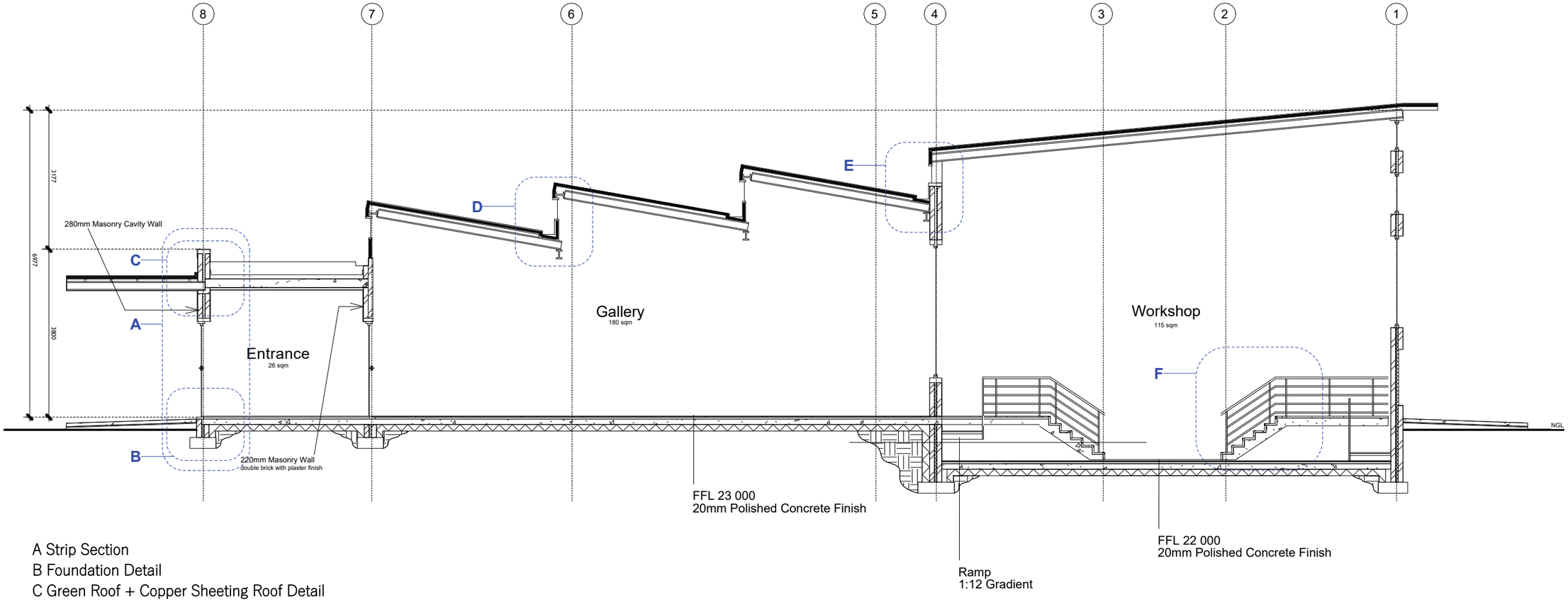
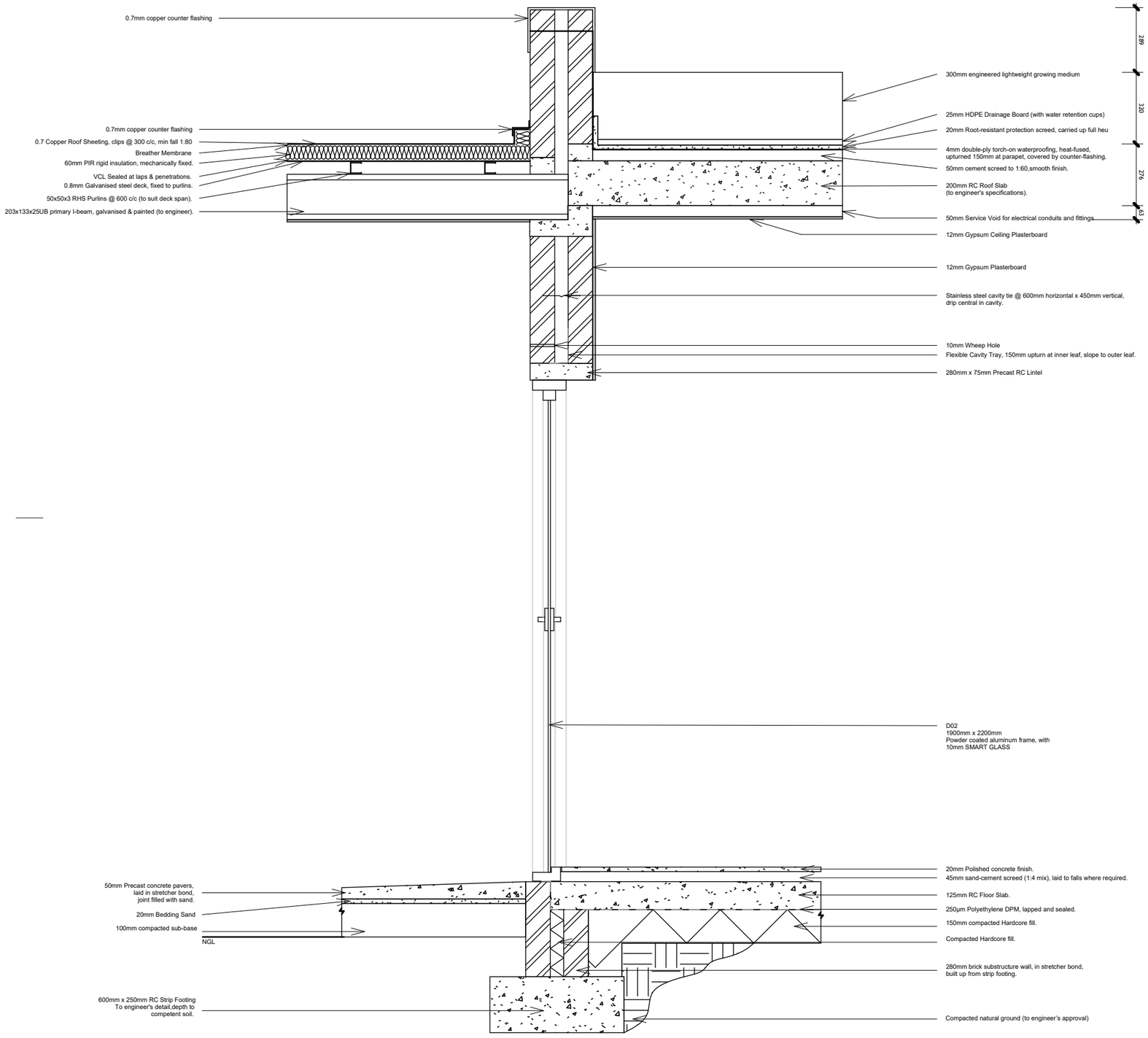
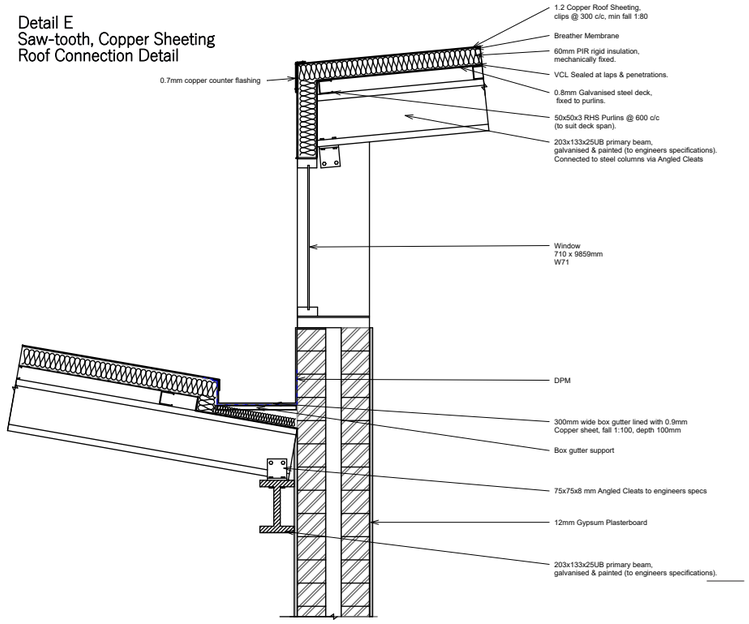


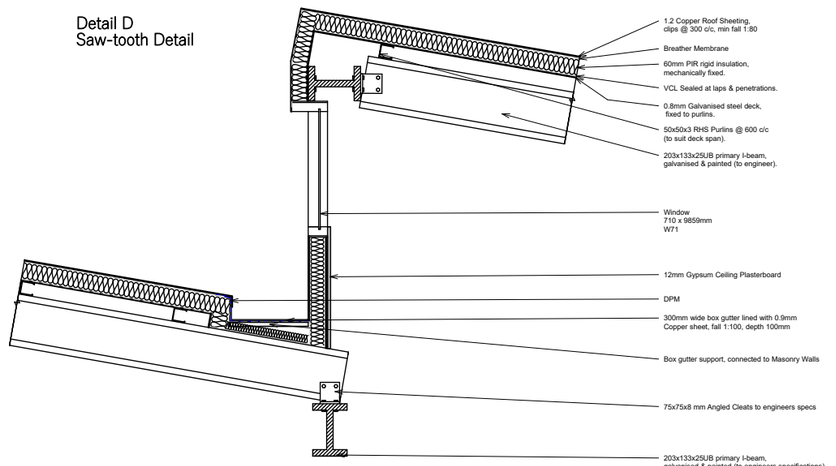
A Strip Section
B Foundation Detail
C Green Roof + Copper Sheeting Roof Detail
D Saw-tooth Roof Detail
E Saw-tooth + Copper Sheeting Roof Connection Detail
F Fire Compliant Stair Detail



Detail E
Saw-tooth, Copper Sheeting Roof Connection Detail



Detail D
Saw-tooth Detail



Gathered Voices

a temporary pavilion for a biennale titled: platform for the unheard

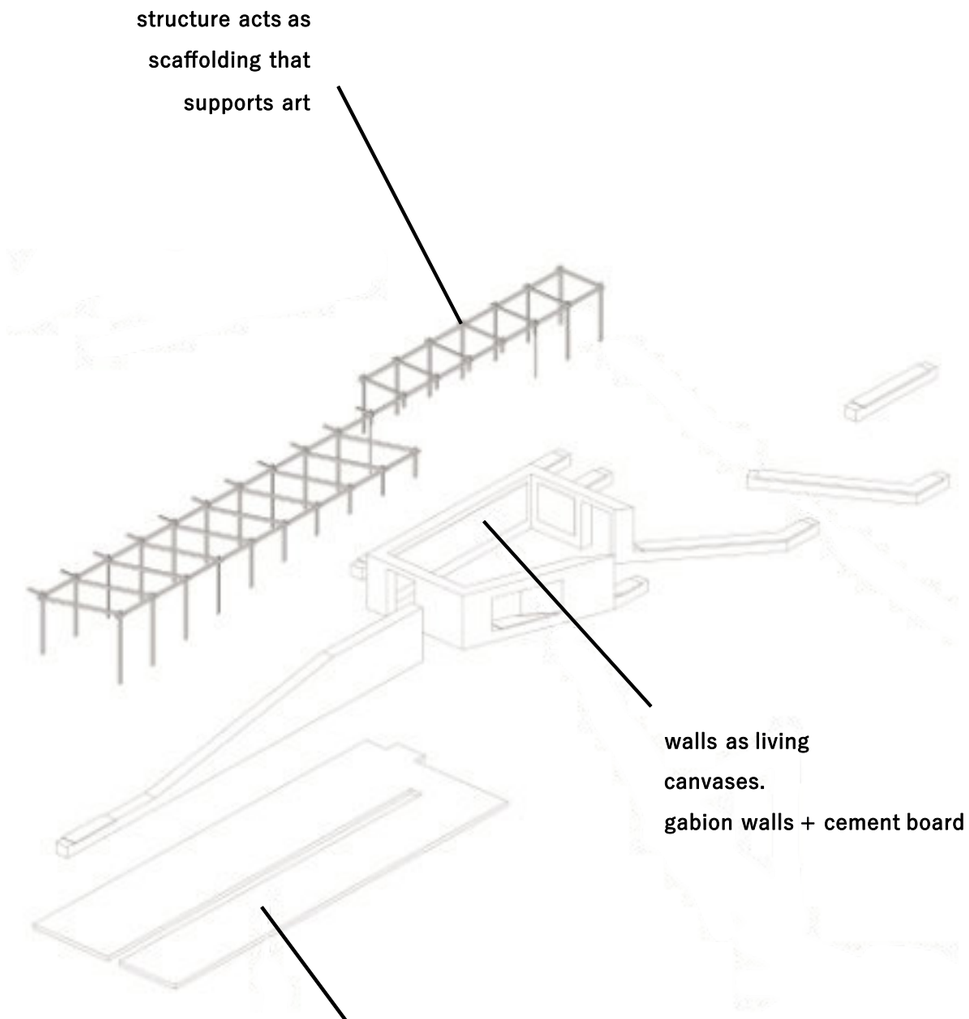
Project Type Temporary Pavilion

Location Sea Point Promenade

Year 2025

Software Revit
Lumion
Procreate

About Designed as a platform for unheard voices, to showcase art, hold discussions and encourage public gathering. Built from gabion walls, timber decking, and a light weight scaffold-like structure for hanging artwork - the pavilion can be assembled and removed without damaging the promenade.


structure acts as
scaffolding that
supports art
walls as living
canvases.
gabion walls + cement board
a possible venue for
discussions and a central
gathering point

Crèche

a preschool informed by Montessori and Waldorf philosophies

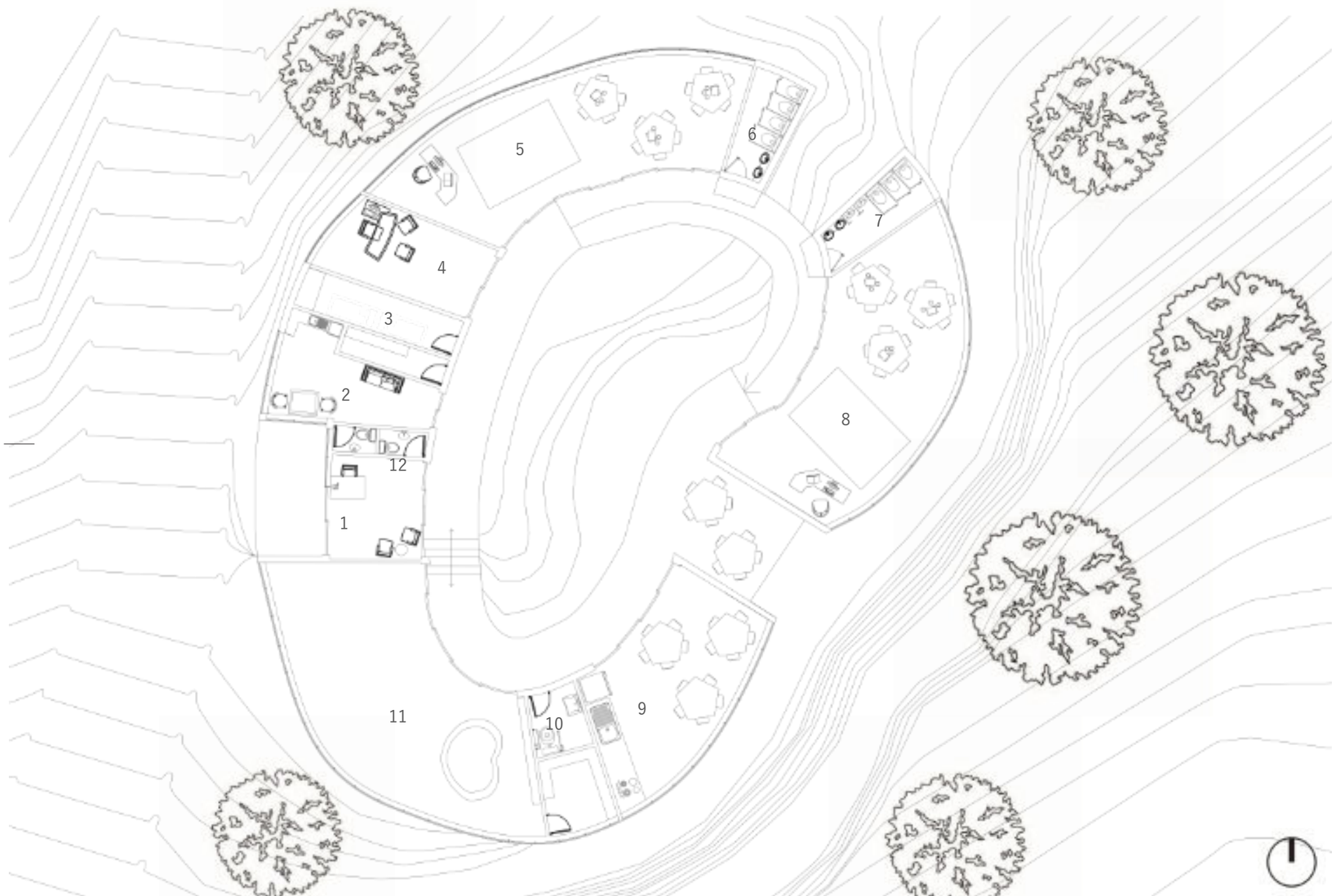
Project Type Crèche

Location Deer Park, Vredehoek

Year 2024

Software Autocad
Affinity

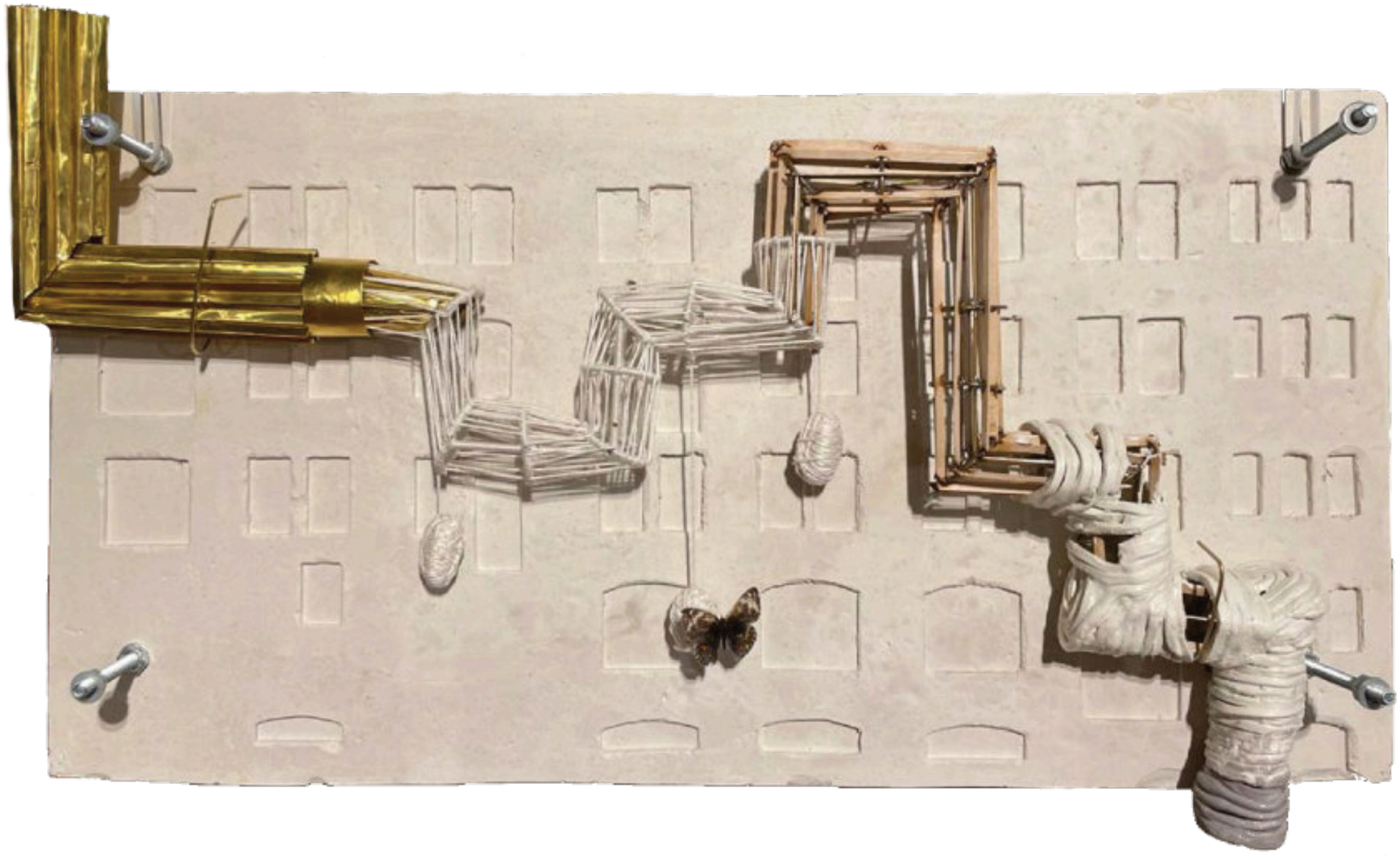
About This Project was a preschool informed by Montessori and Waldorf educational philosophies, exploring how architecture can support children's independence, creativity and learning. The design lead to developing a detailed topographic model made from over 40 contour layers.



1 Entrance
2 Staff Break-room
3 Storage
4 Office
5 Classroom 1
6 Girls Bathroom
7 Boys Bathroom
8 Classroom 2
9 Cafeteria
10 Accessible WC
11 Play area
12 WC

Op-o-site

experimental habitat for a imagined species



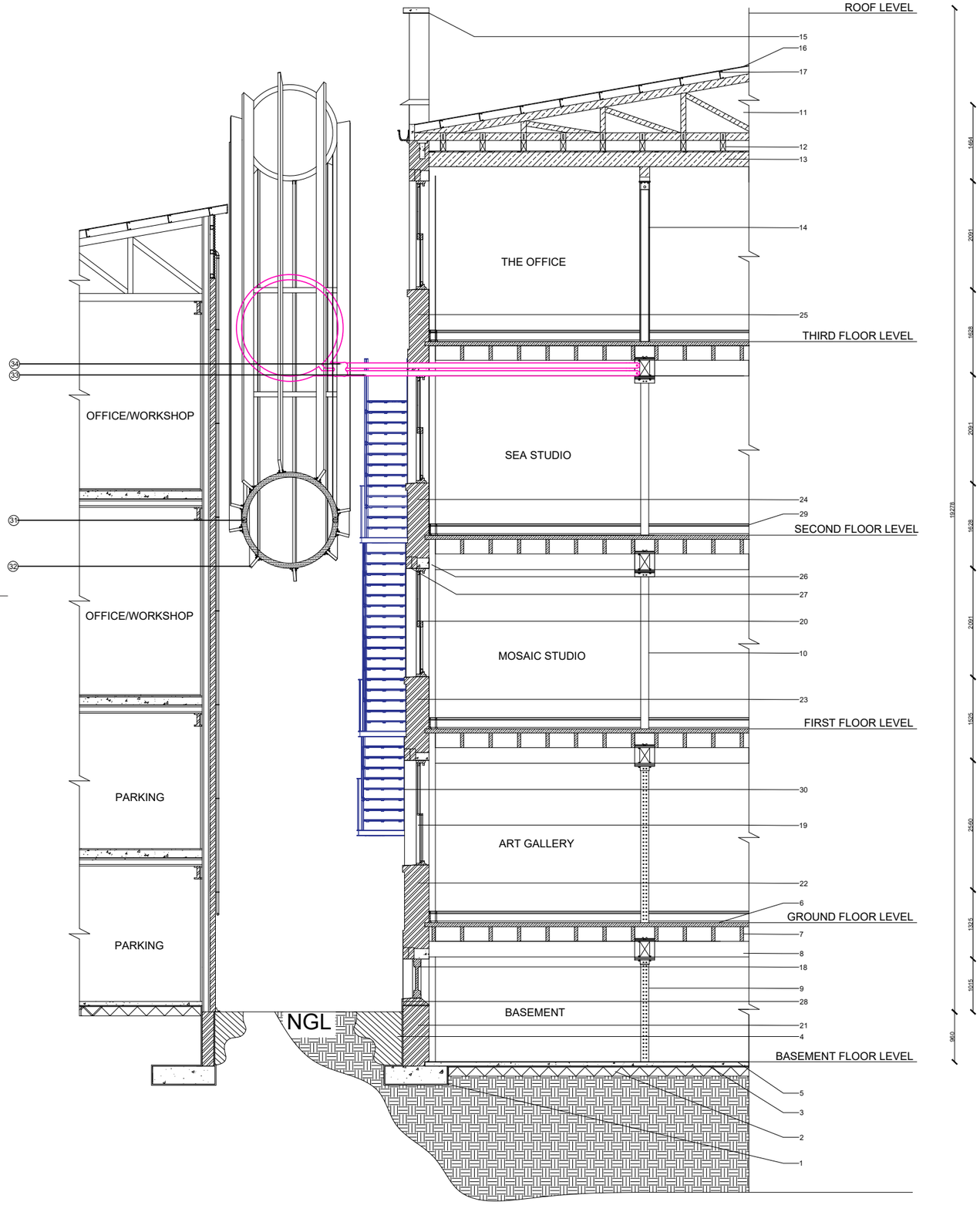
Project Type Abstract Habitat

Location 25 Commercial Street, Cape Town

Year 2024

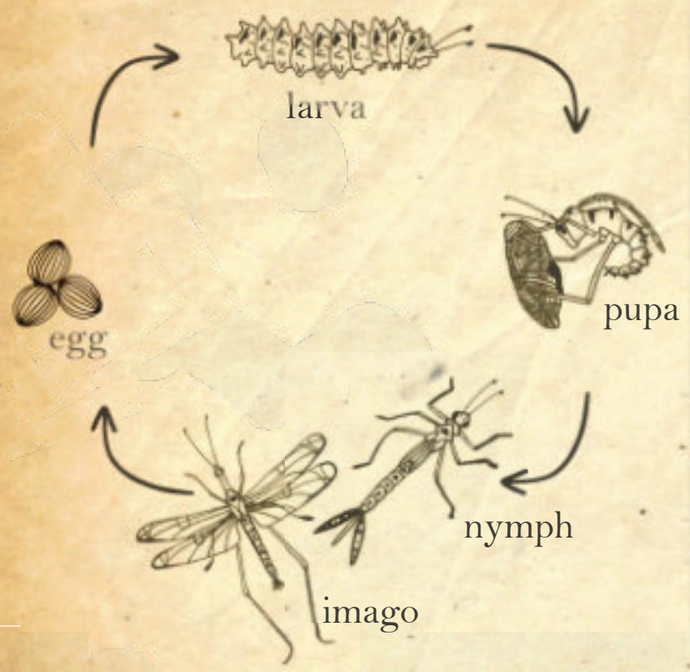
Software Autocad
Affinity

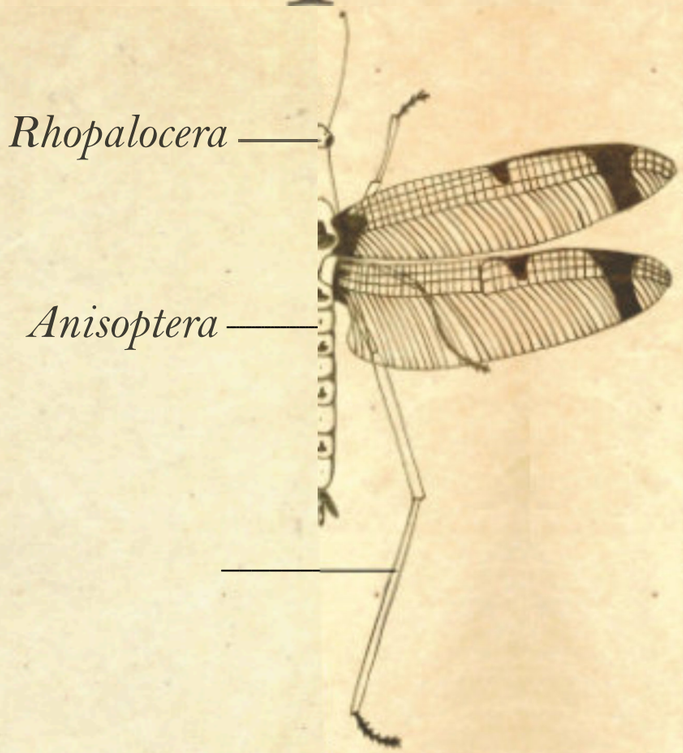
About Designed as an experimental habitat for an imagined species, attached to a school building. Through workshops in timber, clay, steel and fabric, the project introduced me to AutoCAD, laser cutting, soldering, plaster casting, and various other physical prototyping.



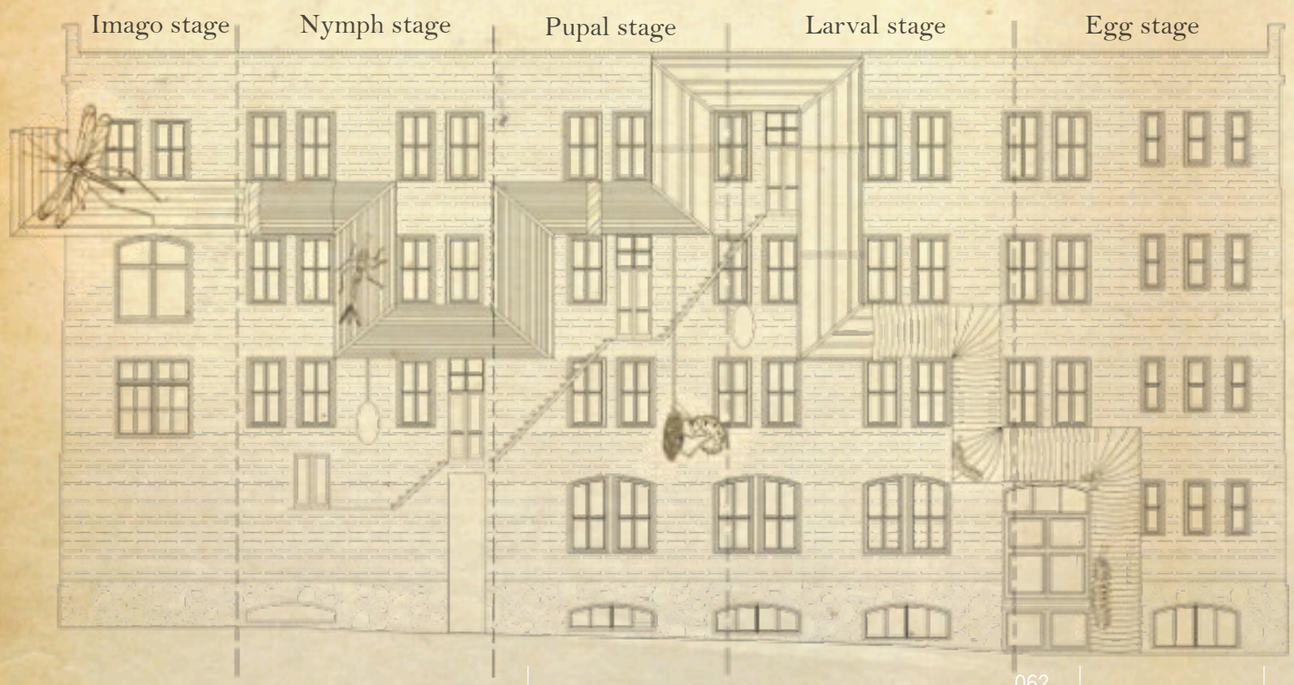
SMIFF table						
Number	Object	Size (mm)	Material	Instance	Finish	Fixing
Foundation						
1	Concrete Foundation Slab	To engineers Specifications	High-strength concrete mix			To engineers specifications
2	Hardcore	To engineers Specifications				To engineers specifications
3	Damp proof membrane		HDPE (High-Density Polyethylene)	Underneath concrete floor		
4	French Drain					
Floors						
5	Basement concrete slab	100	High-strength concrete mix		Troweled	265 mm reinforcement bar
6	Ground, 1st, 2nd and 3rd floor.	150 x 30	Oregon Pine, PAR floorboards		Warm wall nut wood varnish	65mm cut steel nails, 850mm apart
7	Floor joists	270 x 80	Oregon Pine	Every 470mm - Parallel to length of building	Sanded with 80 grip paper	Nailed onto support beams, with 100mm steal-cut nails
8	Support beams	290 x 245 x 3750	Oregon Pine	4580mm - Parallel to width of building	Sanded with 80 grip paper	Supported by I-beam
Beams						
9	I Beam (Parallel Flange)	Length - difference between finished floor levels 160 x 82	Steel	4580mm	Black Paint	50mm x m8 Bolts
10	I Beam (Parallel Flange)	Length - difference between finished floor levels 160 x 82	Steel	4580mm	White Paint	50mm x m8 Bolts
Roof						
11	Roof truss		Oregon Pine	Howe truss to engineers specifications	Sanded with 80 grip paper	
12	Roof joists	270x80	Oregon Pine	Every 470mm - Parallel to length of building	Sanded with 80 grip paper	Nailed onto support beams, with 100mm steal-cut nails
13	Support beams	220 x 150 x 3750	Oregon Pine	4580mm - Parallel to width of building	Sanded with 80 grip paper	Supported by I beam
14	I beams	Length - difference between finished floor levels 160 x 82	Steel	4580mm	White Paint	50mm x m8 Bolts
15	Concrete Parapet		High-strength concrete mix		Plaster and Blue Paint	
16	IBR roof sheeting					Fixed to Purlins
17	Purlins	75 x 50	Galvanised steel	Every 470mm - Parallel to length of building		
Windows						
18	Basement Window	995 x 160	Wood	Repeated as seen in elevation	White paint	Tack nails
19	Ground Floor Window	2550 x 180	Wood	Repeated as seen in elevation	White paint	Tack nails
20	First, Second and Third floor window	2080 x 210	Wood	Repeated as seen in elevation	White paint	Tack nails
Walls						
21	Basement	500 thick	Lime mortar brick	Dutch bond	Bagged wall and blue paint	10mm mortar joint
22	Ground floor	470 thick	Lime mortar brick	Dutch bond	Bagged wall and blue paint	10mm mortar joint
23	First floor	440 thick	Lime mortar brick	Dutch bond	Bagged wall and blue paint	10mm mortar joint
24	Second floor	410 thick	Lime mortar brick	Dutch bond	Bagged wall and blue paint	10mm mortar joint
25	Third floor	380 thick	Lime mortar brick	Dutch bond	Bagged wall and blue paint	10mm mortar joint
26	Lintel	300	Cast in-situ, concrete	Above windows		
27	Lintel	106 x 222 x 72	Brick	Above windows, placed horizontally	Forms part of bagged wall, painted blue	
28	Stone base	1300 x 850 x 700	Pop hammered Granite			
29	Skirting	250 x 70	Timber	Skirting	Painted white	Fixed to wall
Stairs						
30	Stairs		Galvanised steel	Tred, rise	Blue paint	Attached to stretchers with steel pipes
Op-o-site structure						
31	Steel frame	Diameter: 10	Galvanised steel	Cold rolled round tubes		Steel frame welded together
32	Wood beams	250 x 50	Oregon Pine	Radially attached to steel circular frames	Clear wood varnish (ex. WOOD COAT)	50x50mm steel brackets M16 bolts
33	I Beams	160 x 82	Galvanised steel			Attached to 2nd-floor steel I-beams, using M18 bolts.
34	Steel Cleats	250 x 200	Galvanised steel	Attached to I-beams		Fixed to I-beams using M18 bolts.

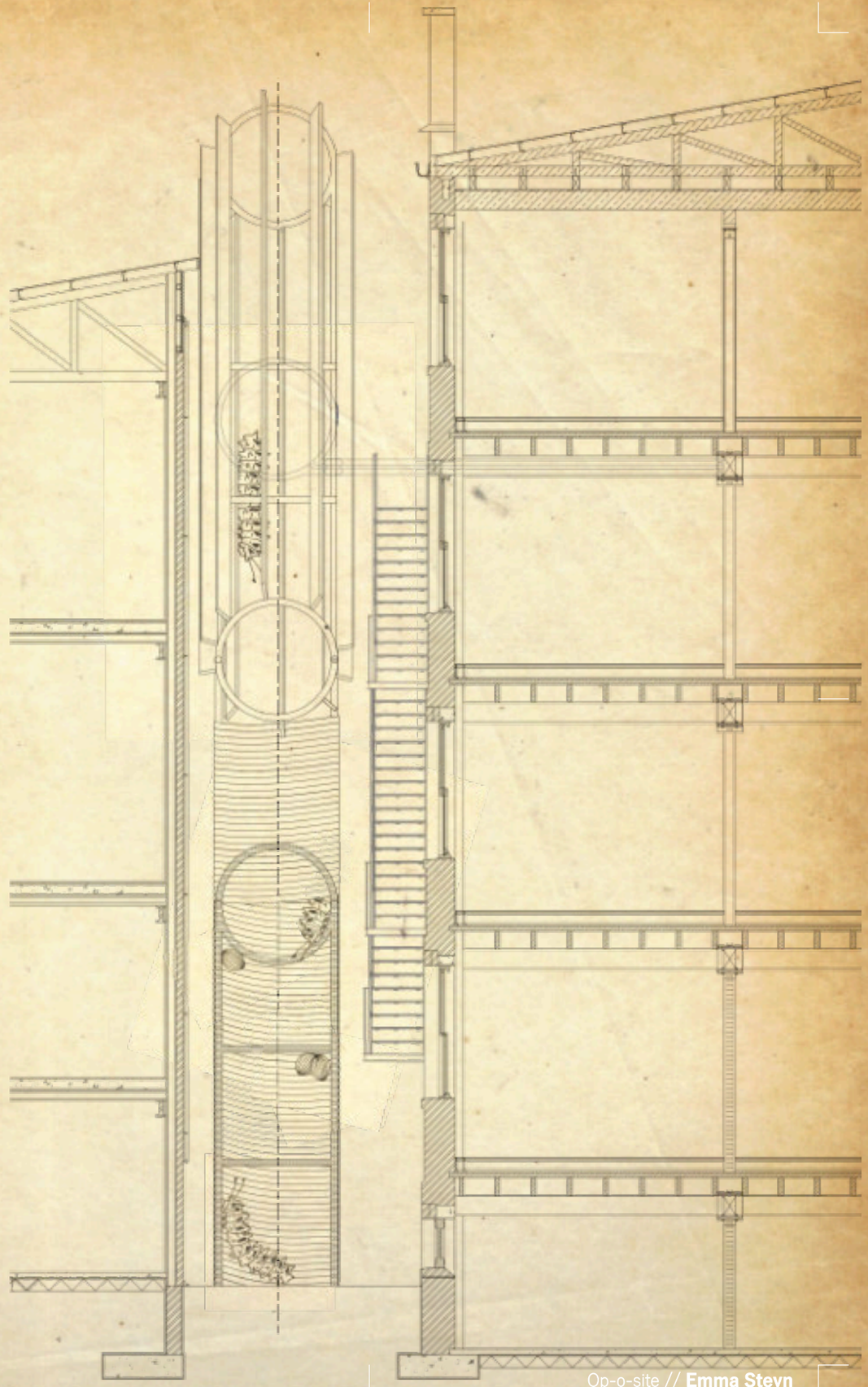
L. Papilomorphis





Metamorphic cycle





Op-o-site // **Emma Steyn**